

Chapter-2: The dot and cross products

Dot product

Definition:

Geometrical interpretation/Explanation of dot product:

Q-13.

Exercise- 63, 64, 65, 67, 84

Physical interpretation/Explanation of dot product:

Q-17.

Exercise- 69

Cross product

Definition:

Geometrical interpretation/Explanation of cross product:

Q-30.

Exercise- 82

Q-32.

Exercise-84

Physical interpretation/Explanation of cross product:

Q-35. Find an expression for the moment of a force $\vec{F}$ about a point P .

Exercise- 86

Triple product

Q-37. Show that $\vec{A} \cdot (\vec{B} \times \vec{C})$ is in absolute value equal to the volume of a parallelepiped with edges $\vec{A}$, $\vec{B}$ and $\vec{C}$.

Exercise- 43, 92

Volume of a tetrahedron:

Q-44. Prove that a necessary and sufficient condition for the vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ to be coplanar is that $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$.

Equation of the plane

Q-18-19. Find an equation for the plane perpendicular to the vector $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and passing through the terminal point of the vector $\vec{B} = \hat{i} + 5\hat{j} + 3\hat{k}$. Find also the distance from the origin to the plane.

Exercise- 77